ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

TYPES OF LANGUAGE MODELS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

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What is Natural Language Processing (NLP)?

N-gram Language Models:

- Markov models
- Likelihood and perplexity
- Smoothing

What are language models and what are they used for?

Text data

Language Model

(1) Evaluate existing text

(2) Generate new text

EXAMPLE: SPEECH RECOGNITION

Speech audio

"Night begins in one hour."

Acoustic model

"Knight begins in one our." X

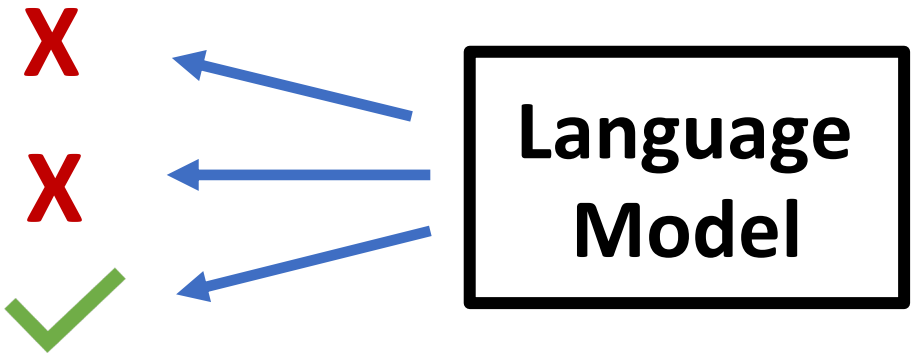
"Knight beg in sin one our." X

"Night begins in one hour." ✓

Language Model

Language models are also useful for:

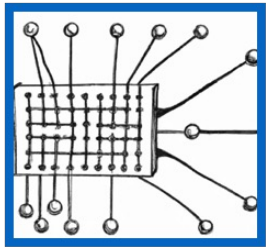
- Machine translation
- Handwriting recognition
- Optical character recognition
- Document summarization
- Dialog generation
- Linguistic decipherment
- Text style transfer
- More…



EXAMPLE: TEXT GENERATION

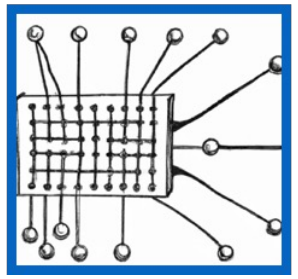
All the text...

Neural Language Model



Prompt

`The cat's...`



Output

`The cat's meow`

EXAMPLE: TEXT GENERATION

Prompt

Write a creative ad for the following product to run on Facebook aimed at parents:

Product: Learning Room is a virtual environment to help students from kindergarten to high school excel in school.

Output

Are you looking for a way to help your child excel in school? Look no further than Learning Room! Our virtual environment is designed to help students from kindergarten to high school learn and grow. With our help, your child will be able to get ahead in school and achieve their academic goals.

source: https://beta.openai.com/examples

EXAMPLE: TEXT SUMMARIZATION

Prompt

A neutron star is the collapsed core of a massive supergiant star, which had a total mass of between 10 and 25 solar masses, possibly more if the star was especially metal-rich. [1] Neutron stars are the smallest and densest stellar objects, excluding black holes and hypothetical white holes, quark stars, and strange stars. [2] Neutron stars have a radius on the order of 10 kilometres (6.2 mi) and a mass of about 1.4 solar masses. [3] They result from the supernova explosion of a massive star, combined with gravitational collapse, that compresses the core past white dwarf star density to that of atomic nuclei.

Tl;dr

Output

A neutron star is the collapsed core of a massive supergiant star. These ultra-dense objects are incredibly fascinating due to their strange properties and their potential for phenomena such as extreme gravitational forces and a strong magnetic field.

source: https://beta.openai.com/examples

EXAMPLE: CODE GENERATION

Prompt

Get the current price of Bitcoin using Coindesk's API and display it as giant red letters in the middle of a webpage

Output

```
/* Get the current price of Bitcoin using Coindesk's API and
display it as giant red letters in the middle of a webpage */
var xhr = new XMLHttpRequest();
xhr.open('GET','https://api.coindesk.com/vi/bpi/currentprice.json',
true);
xhr.onload = function() {
   if (xhr.status === 200) {
      var data = JSON.parse (xhr.responseText);
      var bitcoinPrice = document.createElement('div');
      bitcoinPrice.innerHTML= data.bpi.USD.rate;
      bitcoinPrice.style.fontSize = '100px';
      bitcoinPrice.style.color = 'red';
   document.body.appendChild(bitcoinPrice);
}
};
xhr.send();
```

source: https://betterprogramming.pub/

A language model is a probability distribution over sequences of words (sentences):

$$P(w) = P(w_1 \dots w_n)$$

- What's w? (closed vs open vocabulary)
- What's n? (must sum to one over all lengths)
- Can have rich structure or be linguistically naive

Why language models?

- Want "good" (e.g. fluent) sentences to have high probability
- Evaluate sentences via their probability
- Generate sentences by sampling from the distribution

Types of language models:

- **N-Gram language models** (this lecture) – simple and fast, limited power
- **Neural language models** – powerful, more complex, slower to train, slower to use in some applications
- **Conditional language models** – conditions on an input, used for applications like machine translation and summarization